DECEMBER	TOPIC
11	TEST OF HYPOTHESIS (1/2)

HYPOTHESIS ETYMOLOGY

ὑποτίθημι

hupotíthēmi

ὑπό + τίθημι

(hupó, “below”) + (títhēmi, “I put, place”)

«I set suggest»

«Varsayıyorum»

INTRODUCTION

A hypothesis test is similar to a scientific method.

A scientist observes the nature and establishes a theory and tests the theory against the reality.

INTRODUCTION

In statistics, there is a theory for the population parameters.

We draw a sample from the population and theory is tested based on the information contained in this sample.

INTRODUCTION

If the result is not compatible with the theory, it is rejected.

If it is compatible, the theory is either correct or there is no difference between the true.

INTRODUCTION

A manufacturer of emergency equipment asserts that a respirator that it makes delivers pure air for 75 minutes on average. A government regulatory agency is charged with testing such claims, in this case to verify that the average time is not less than 75 minutes. To do so it would select a random sample of respirators, compute the mean time that they deliver pure air, and compare that mean to the asserted time 75 minutes.

INTRODUCTION

In the sampling the goal has been to estimate a population parameter. **(RECALL: Estimation Lecture)**

But the sampling done by the government agency has a somewhat different objective, not so much to estimate the population mean μ as to test an assertion about it.

INTRODUCTION

Whether it is as large as 75 or not.

The agency is not necessarily interested in the actual value of μ, just whether it is as claimed.

Their sampling is done to perform:

a test of hypotheses.

INTRODUCTION

A hypothesis about the value of a population parameter is an assertion about its value.

We will be concerned with testing the truth of two competing hypotheses, only one of which can be true.

INTRODUCTION

For example, suppose that the average grade of the ore deposit is greater than 5%.

The hypothesis is that the average grade is greater than 5%. If we do not believe that the hypotheses is correct, than we claim that the average grade is less than 5%.

DEFINITIONS

The null hypothesis, denoted H_o, is the statement about the population parameter that is assumed to be true unless there is convincing evidence to the contrary.

The alternative hypothesis, denoted H_a, is a statement about the population parameter that is contradictory to the null hypothesis, and is accepted as true only if there is convincing evidence in favor of it.

DEFINITIONS

Hypothesis testing is a statistical procedure in which a choice is made between a null hypothesis and an alternative hypothesis based on information in a sample.

Rejection region determines the values of the test statistics that null hypothesis is rejected.

DEFINITIONS

The end result of a hypotheses testing procedure is a choice of one of the following two possible conclusions:

Reject H_o ($\therefore$ accept H_a)

Fail to reject H_o ($\therefore$ fail to accept H_a).

DEFINITIONS

The null hypothesis typically represents the status quo, or what has historically been true.

In the example of the respirators, we would believe the claim of the manufacturer unless there is reason not to do so, so the null hypotheses is H_o:μ=75.

DEFINITIONS

The alternative hypothesis in the example is the contradictory statement H_a:$\mu<75$.

The null hypothesis will always be an assertion containing an equals sign, but depending on the situation the alternative hypothesis can have any one of three forms: with the symbol **"<"** as in the example just discussed, with the symbol **">"** or with the symbol **"≠"**

The first step in hypothesis testing is to identify the null and alternative hypotheses

EXAMPLES

We claim that the average grade is less than 5%.

Null hypothesis = H_o:$\mu<5$

Alternative hypothesis = H_a:$\mu=5$ or $\mu>5$

EXAMPLES

A publisher of college textbooks claims that the average price of all hardbound college textbooks is \$127.50. A student group believes that the actual mean is higher and wishes to test their belief. State the relevant null and alternative hypotheses.

Null hypothesis = H_o: $\mu=127.5$

Alternative hypothesis = H_a: $\mu>127.5$

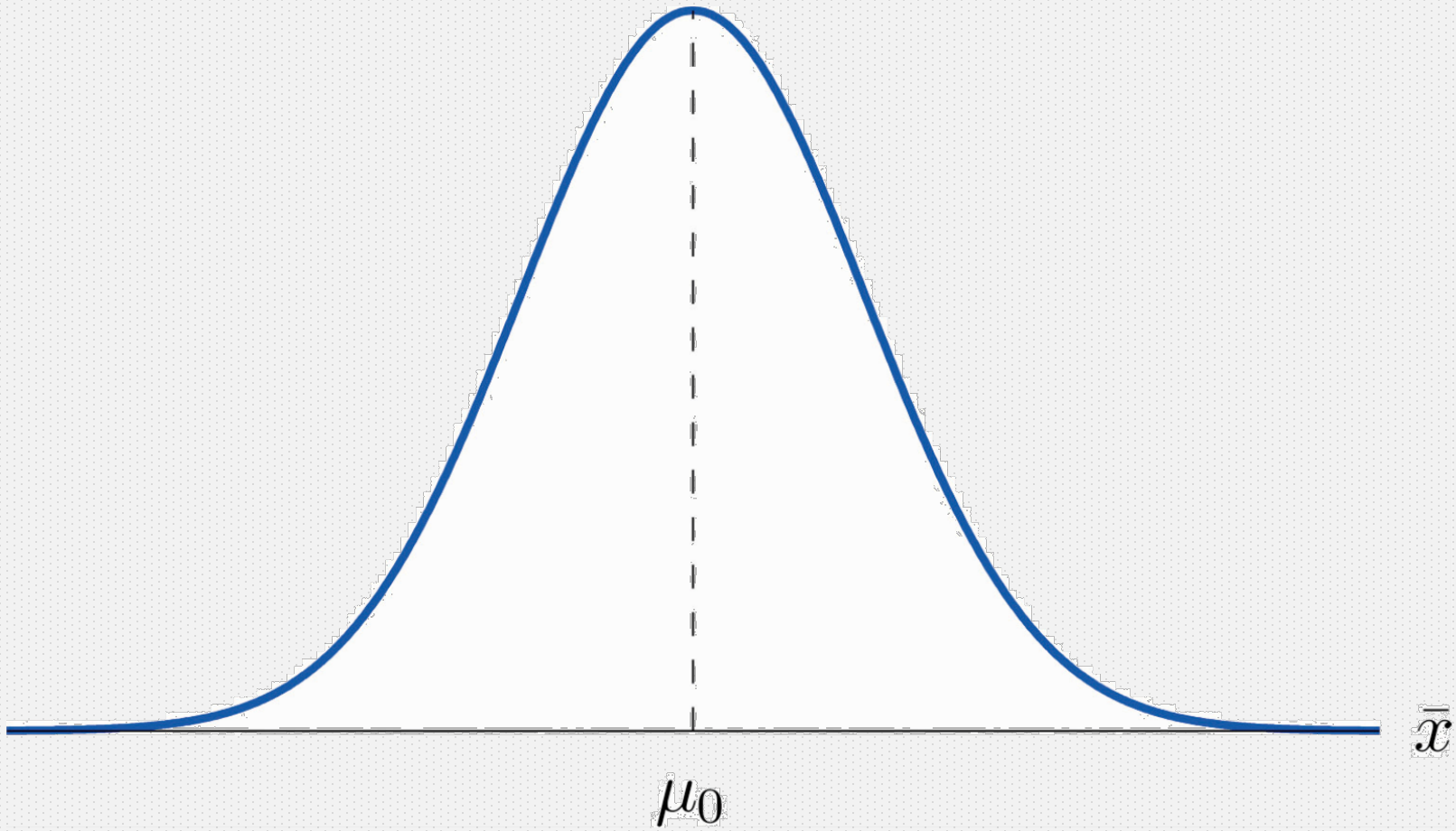
REJECTION

The criterion for judging between H_o and Ha based on the sample data is: if the value of $\bar{x}$ would be highly unlikely to occur if H_o were true, but favors the truth of H_a, then we reject H_o in favor of H_a. Otherwise we do reject H_o.

REJECTION

Suppose $\bar{x}$ follows a normal distribution, when the null hypothesis is true the density function for the sample mean $\bar{x}$ must be as is a bell curve centered at μ_0.

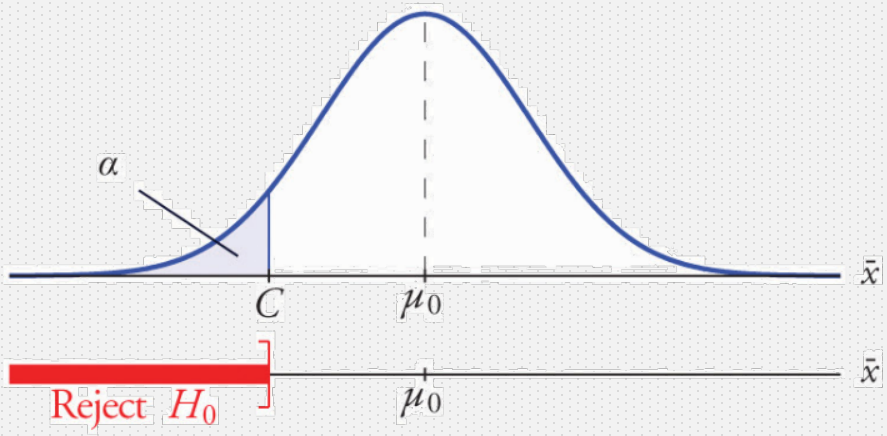
Thus, if Ho is true then $\bar{x}$ is likely to take a value near μ_0 and is unlikely to take values far away.


$\bar{x}$
μ_0

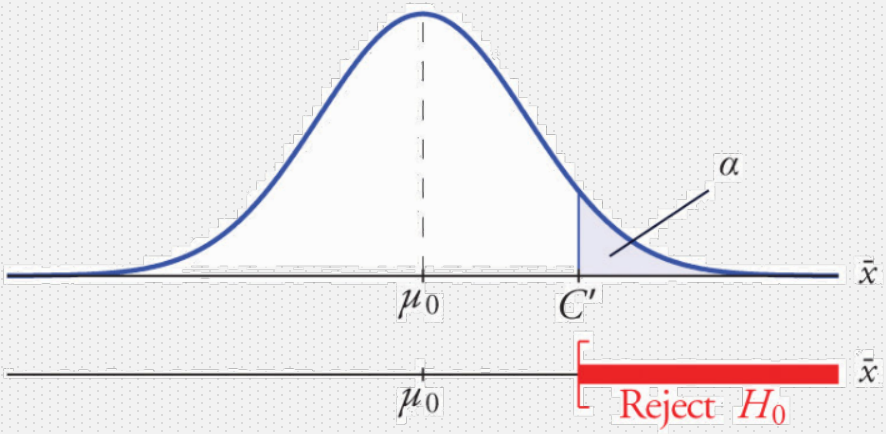
REJECTION

1. If H_a has the form H_a: $\mu < \mu_0$ then reject H_o if $\bar{x}$ is far to the left of μ_0
2. If H_a has the form H_a: $\mu > \mu_0$ then reject H_o if $\bar{x}$ is far to the right of μ_0
3. If H_a has the form H_a: $\mu \neq \mu_0$ then reject H_o if $\bar{x}$ is far away from μ_0 in either direction.

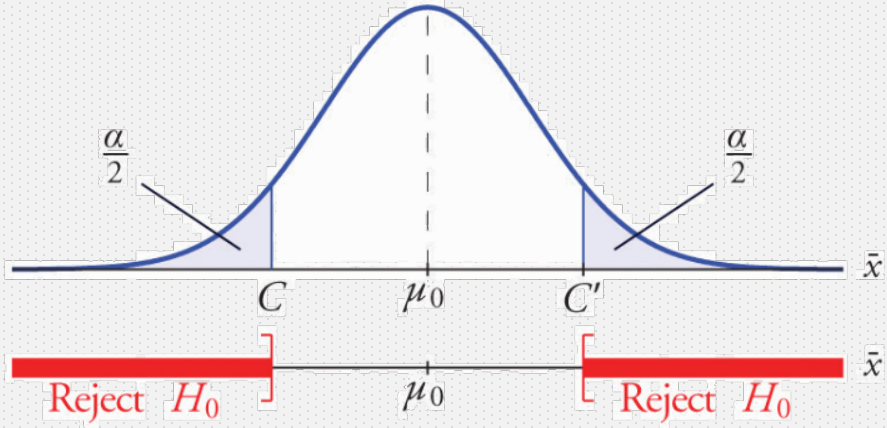
$H_a : \mu < \mu_0$



$H_a : \mu > \mu_0$



$H_a : \mu \neq \mu_0$



DECISION PROCEDURE

If H_a has the form $\mu \neq \mu_0$ the test is called a **two-tailed test.**

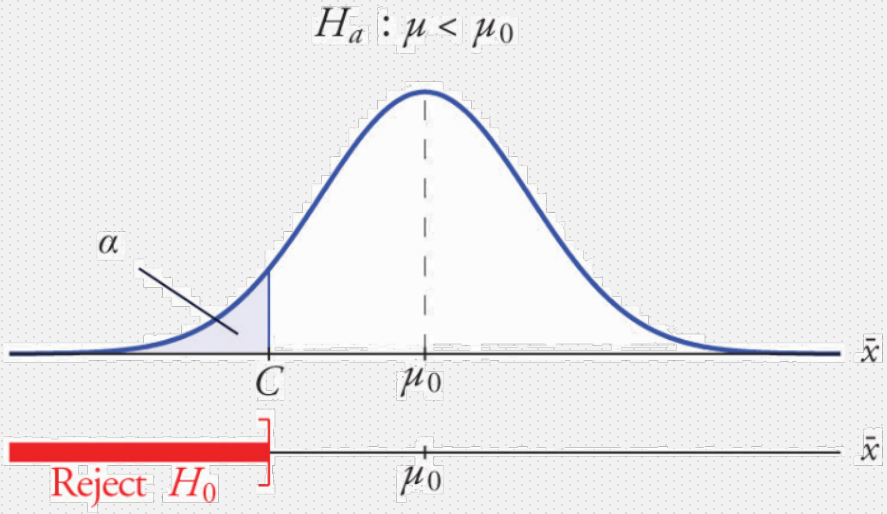
If H_a has the form $\mu < \mu_0$ the test is called a **left-tailed test.**

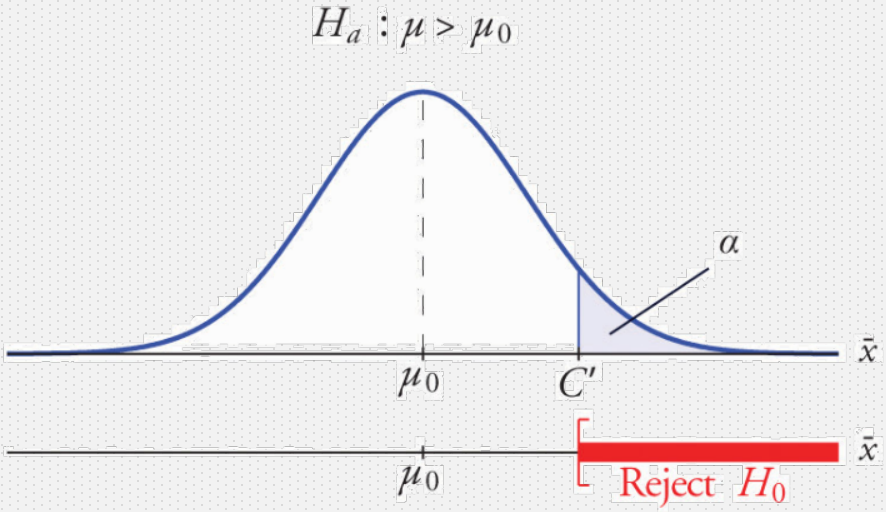
If H_a has the form $\mu > \mu_0$ the test is called a **right-tailed test.**

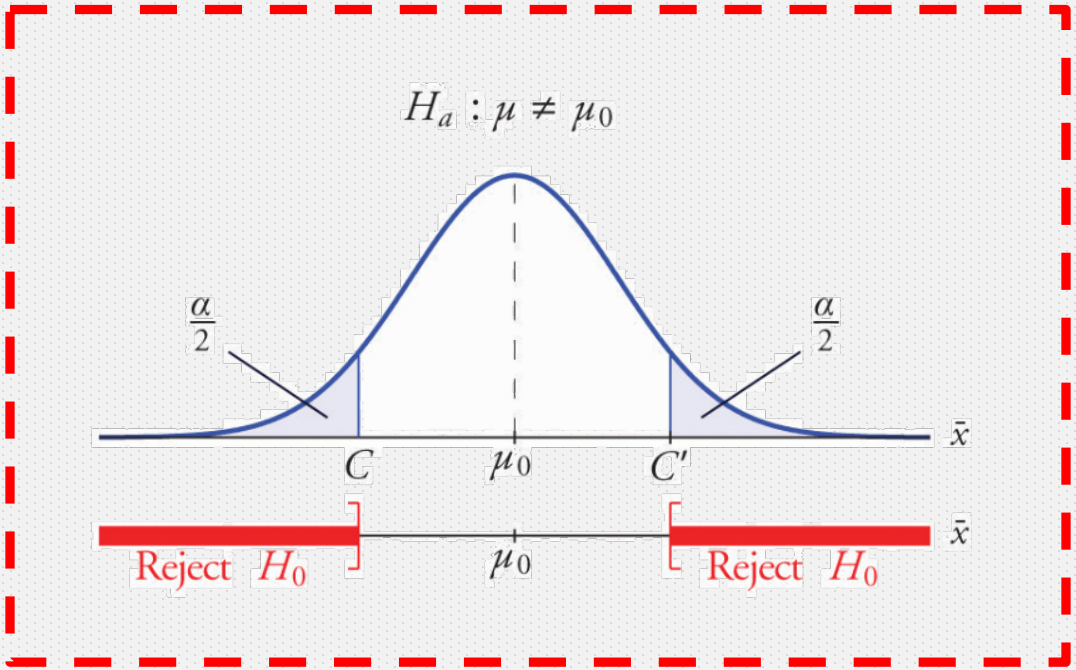
EXAMPLE

The recipe for a bakery item is designed to result in a product that contains 8 grams of fat per serving. The quality control department samples the product periodically to insure that the production process is working as designed. State the relevant null and alternative hypotheses. The population is normally distributed with $\sigma = 0.15$

Construct the rejection region of 90% CI for 5 samples.


$H_a : \mu < \mu_0$
α
C
μ_0
$\bar{x}$
Reject H_0
μ_0
$\bar{x}$


$H_a : \mu > \mu_0$
α
μ_0
C'
$\bar{x}$
μ_0
Reject H_0
$\bar{x}$


$H_a : \mu \neq \mu_0$
$\frac{\alpha}{2}$
$\frac{\alpha}{2}$
C
μ_0
C'
$\bar{x}$
Reject H_0
μ_0
Reject H_0
$\bar{x}$

$\mu \neq \mu_0$

CRITICAL VALUES OF $t_{a/2}$

df	0.250	0.200	0.150	0.100	0.050	0.025	0.010	0.005	0.001	0.0005
1	1.000	1.376	1.963	3.078	6.314	12.710	31.820	63.660	318.310	636.620
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
.	.	.	.	.	.	.	.	.	.	.
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29	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
∞	0.842	1.036	1.282	1.645	1.960	2.326	2.576	2.807	3.090	3.291

Null hypothesis = H_o : μ=8

Alternative hypothesis = H_a: μ ≠ 8

$$\bar{x} - t_{\alpha/2}\frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{\alpha/2}\frac{s}{\sqrt{n}}$$

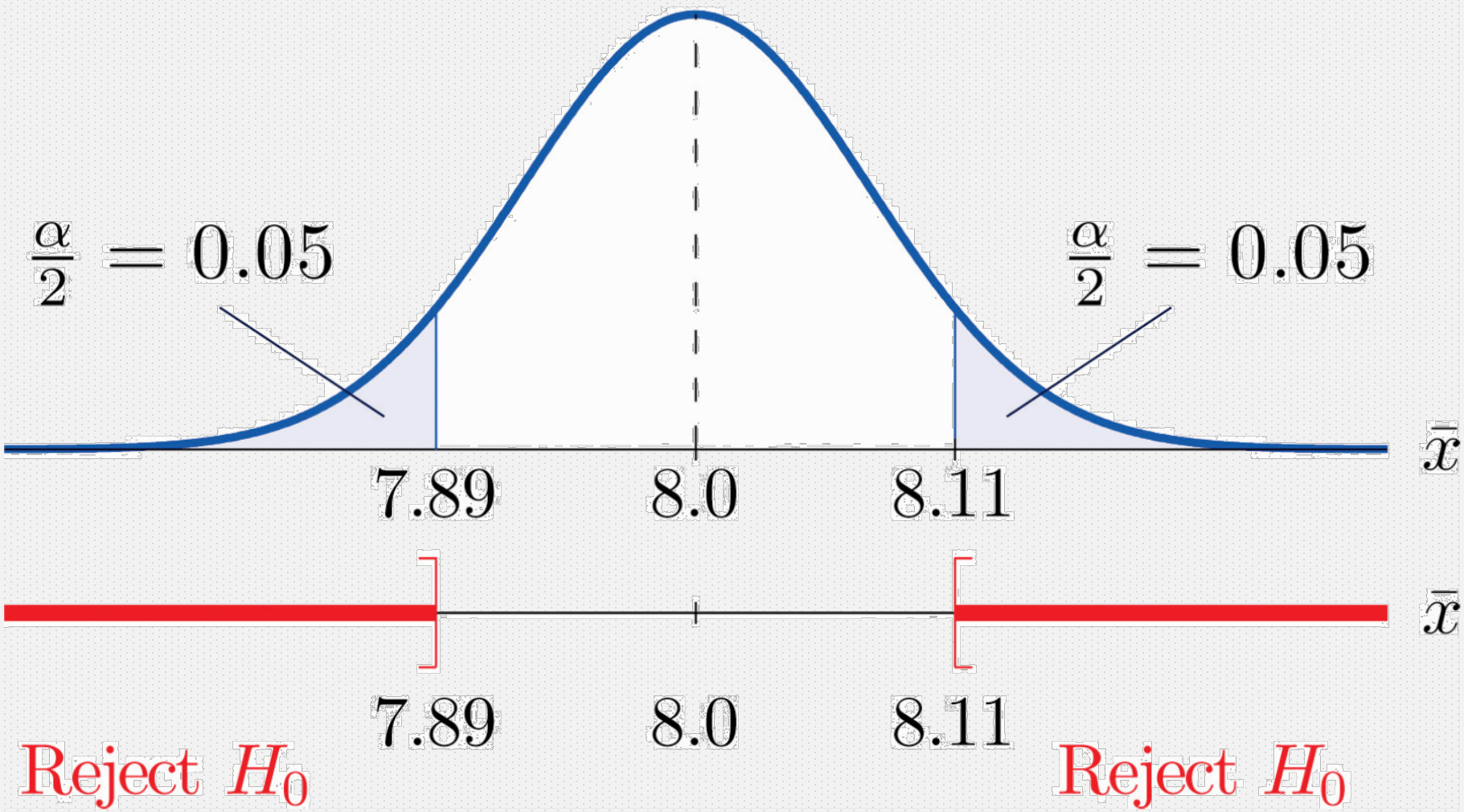
$$\bar{\sigma}_x = \frac{0.15}{\sqrt{5}} = 0.067$$

$$t_{\alpha/2} = 1.645$$

$$C = 8 + 0.067 \times 1.645 = 7.89$$

$$C' = 8 - 0.067 \times 1.645 = 8.11$$

$$H_a : \mu \neq 8.0$$


$\frac{\alpha}{2} = 0.05$
$\frac{\alpha}{2} = 0.05$
$\bar{x}$
7.89
8.0
8.11
$\bar{x}$
7.89
8.0
8.11
Reject H_0
Reject H_0

DECISION PROCEDURE

Take a sample of size 5 and compute the sample mean $\bar{x}$. If $\bar{x}$ is either 7.89 grams or less or 8.11 grams or more then **reject the hypothesis** that the average amount of fat in all servings of the product is 8.0 grams.

Otherwise do not reject the hypothesis that the average amount is 8.0 grams.